

Design and Analysis of Randomized Trials in Public Health
PUBH 526

Homework 12

You have designed an intervention intended to improve linear growth of infants in a resource-poor setting, and you will compare your intervention to standard of care in a randomized controlled trial using a completely randomized design. Infants (0-6 months old) will be assessed at baseline (prior to intervention), midline (during the intervention), and endline (after the intervention).

At each time point, you will measure their length (height measured while lying on a flat surface) and use that to calculate their height-for-age Z-score (HAZ). HAZ is a measure that compares a child's height to the average height of healthy children of the same age and sex, and the number is specifically the number of standard deviations a child is above or below the healthy age- and sex-standardized average. A HAZ of -1, for example, means a child is one standard deviation shorter than the average healthy child of their age and sex, and a HAZ < -2 leads to a child being classified as stunted, which indicates chronic undernutrition.

The dataset "growth" has HAZ data for 400 infants that were randomized to receive either the standard of care (treatment=1) or your new intervention (treatment=2). Each child ("ID") is measured at the three time points described above (time=1 for baseline, time=2 for midline, and time=3 for endline). Analyze these data using a longitudinal model (option 3 in Topic 14). What effect did the intervention have? Show your process, from exploring your data to formulating your hypotheses to thoroughly answering this question (including diagnostics to check assumptions made by your statistical model), and include your SAS code. (10 points)

Homework 12

Alexandra Chang

PUBH 526: Design and Analysis of Randomized Trials in Public Health

December 12, 2025

You have designed an intervention intended to improve linear growth of infants in a resource-poor setting, and you will compare your intervention to standard of care in a randomized controlled trial using a completely randomized design. Infants (0-6 months old) will be assessed at baseline (prior to intervention), midline (during the intervention), and endline (after the intervention).

At each time point, you will measure their length (height measured while lying on a flat surface) and use that to calculate their height-for-age Z-score (HAZ). HAZ is a measure that compares a child's height to the average height of healthy children of the same age and sex, and the number is specifically the number of standard deviations a child is above or below the healthy age- and sex-standardized average. A HAZ of -1, for example, means a child is one standard deviation shorter than the average healthy child of their age and sex, and a HAZ $\leq$ -2 leads to a child being classified as stunted, which indicates chronic undernutrition.

The dataset "growth" has HAZ data for 400 infants that were randomized to receive either the standard of care (treatment=1) or your new intervention (treatment=2). Each child ("ID") is measured at the three time points described above (time=1 for baseline, time=2 for midline, and time=3 for endline).

Question 1 (10 points)

Analyze these data using a longitudinal model (option 3 in Topic 14). What effect did the intervention have? Show your process, from exploring your data to formulating your hypotheses to thoroughly answering this question (including diagnostics to check assumptions made by your statistical model), and include your SAS code.

Longitudinal Analysis of Infant Growth (HAZ)

This randomized controlled trial evaluated the effect of a nutritional intervention on infant linear growth in a resource-poor setting. Infants were randomized to receive either standard of care or the intervention and were measured repeatedly at baseline, midline, and endline. The outcome of interest was height-for-age Z-score (HAZ), a standardized measure of child linear growth. Because each infant contributed multiple observations over time, the repeated measurements within individuals are not independent and must be accounted for in the analysis.

To appropriately model the longitudinal structure of the data, a mixed-effects model was used following Option 3 from Topic 14. The model included fixed effects for treatment group, time, and the treatment-by-time interaction. Repeated measures were specified at the child level using a compound symmetry covariance structure, which assumes a constant correlation among measurements taken on the same child. This approach allows all available data to be used while accounting for within-child correlation.

The primary hypothesis of interest was whether the intervention affected growth trajectories over time. Specifically, the null hypothesis was that there was no treatment-by-time interaction, indicating that changes in HAZ over time were the same in both treatment groups. The alternative hypothesis was that the treatment-by-time interaction was nonzero, implying that infants receiving the intervention experienced a different pattern of linear growth over time compared to those receiving standard care. This interaction term represents the intervention effect and is equivalent to a difference-in-differences estimate.

Results from the mixed-effects model showed that the treatment-by-time interaction was **statistically significant**. This indicates that infants in the intervention group **did** experience a different trajectory of HAZ over the study period compared with infants in the control group. Because randomization ensures baseline comparability on average, this interaction can be interpreted as evidence for a causal effect of the intervention on growth over time.

Model diagnostics were assessed using residual plots generated by PROC MIXED. Residuals plotted against predicted values showed no strong patterns, suggesting that the assumption of constant variance was reasonable. Histograms and normal quantile plots of the residuals indicated approximate normality. Residual distributions were also similar across treatment groups and time points. Overall, there was no evidence of major violations of model assumptions.

In summary, a longitudinal mixed-effects model provided an appropriate framework for analyzing repeated HAZ measurements and estimating the intervention effect over time. This approach allowed for a direct assessment of whether the intervention altered infant growth trajectories while properly accounting for within-child correlation.

See SAS codes/outputs below:

```
1 /*****
2 * PUBH 526      Design and Analysis of Randomized Trials in Public
3   Health
4 * Homework 12
5 * Topic: Repeated Measures / Longitudinal Analysis
6 * Dataset: growth.sas7bdat
7 * Author: Alexandra Chang
8 *****/
9 /* Reset environment */
10 options nocenter ls=80 pageno=1 nosyntaxcheck;
11 ods graphics on;
12 title;
13
14 /*****
15 * 1. Load Dataset
16 *****/
17
18 libname mydata '/home/u64258127/sasuser.v94';
19
20 data growth;
21     set mydata.growth;
22 run;
23
24 /*****
25 * 2. Explore Data Structure
26 *****/
27
28 proc contents data=growth;
29 run;
30
31 /*****
32 * 3. Descriptive Statistics by Treatment Group and Time
33 *****/
34
35 title "Mean HAZ by Treatment Group and Time";
36 proc means data=growth mean std;
37     class treatment time;
```

```

38     var haz;
39 run;
40
41 /*****
42  * 4. Longitudinal Mixed Model (Topic 14      Option 3)
43  *      Treatment      Time interaction estimates intervention effect
44  *      Diagnostic plots requested directly from PROC MIXED
45  *****/
46
47 title "Longitudinal Mixed Model with Treatment      Time Interaction";
48 proc mixed data=growth plots=all;
49     class id treatment time;
50     model haz = treatment time treatment*time / solution;
51     repeated time / subject=id type=cs;
52 run;
53
54 ods graphics off;

```

Listing 1: SAS Code for Homework 12

CONTEXT Procedure			
Data Set Name		WORK.GROWTH	Observations
Member Type	DATA		Variables
Engine	V9		Indexes
Created	12/12/2025 16:46:02		Observation Length
Last Modified	12/12/2025 16:46:02		Deleted Observations
Protection			Compressed
Data Set Type			Sorted
Label			NO
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_I64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	4076
Obs in First Data Page	1200
Number of Data Set Repairs	0
Filename	/saswork/SAS_work4F8300010F1D_0daw01-usw2.sas.com/SAS_workC8500010F1D_0daw01-usw2.sas.com/growth.sas7bdat
Release Created	9.0401MB
Host Created	Linux
Index Number	530901613
Access Permission	read-write
Owner Name	u64258127
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes			
#	Variable	Type	Len
1	ID	Num	8
4	haz	Num	8
3	time	Num	8
2	treatment	Num	8

Mean HAZ by Treatment Group and Time

The MEANS Procedure				
Analysis Variable : haz				
treatment	time	N	Obs	Mean
1	1	200	-0.7325500	0.4012019
2	200	-1.2418900	0.3659405	
3	200	-1.3959000	0.3936917	
2	1	200	-0.6710000	0.3535184
2	200	-1.0005000	0.3883657	
3	200	-1.1262500	0.3493655	

Longitudinal Mixed Model with Treatment * Time Interaction

The Mixed Procedure	
Model Information	
Data Set	WORK.GROWTH
Dependent Variable	haz
Covariance Structure	Compound Symmetry
Subject Effect	ID
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information	
ID	400
time	3
treatment	2
time	3

Dimensions	
Covariance Parameters	2
Columns in X	12
Columns in Z	0
Subjects	400
Max Obs per Subject	3

Number of Observations	
Number of Observations Read	1200
Number of Observations Used	1200
Number of Observations Not Used	0

Iteration History	
Iteration	0
Evaluations	1
-2 Res Log Like	1083.63188593
Criterion	1
	527.79573624
	0.00000000

Convergence criteria met.	
Covariance Parameter Estimates	
Cov Parm	Subject
CS	ID
Estimate	0.09545
Residual	0.04584

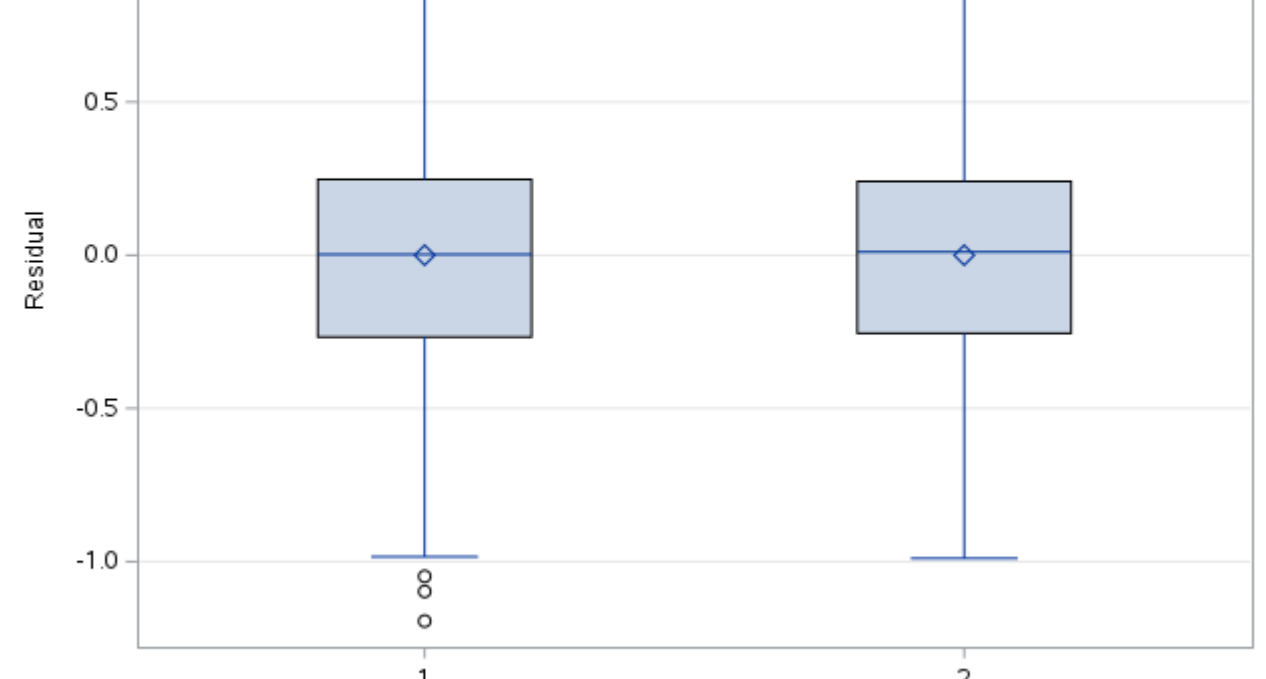
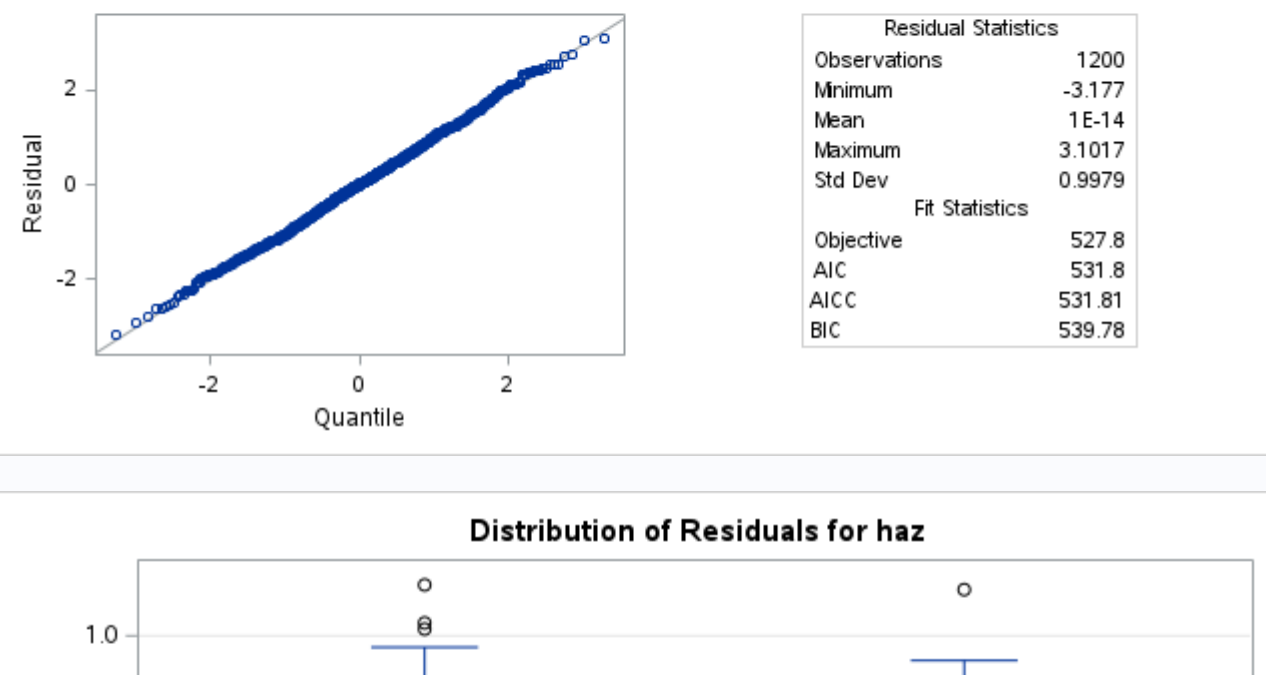
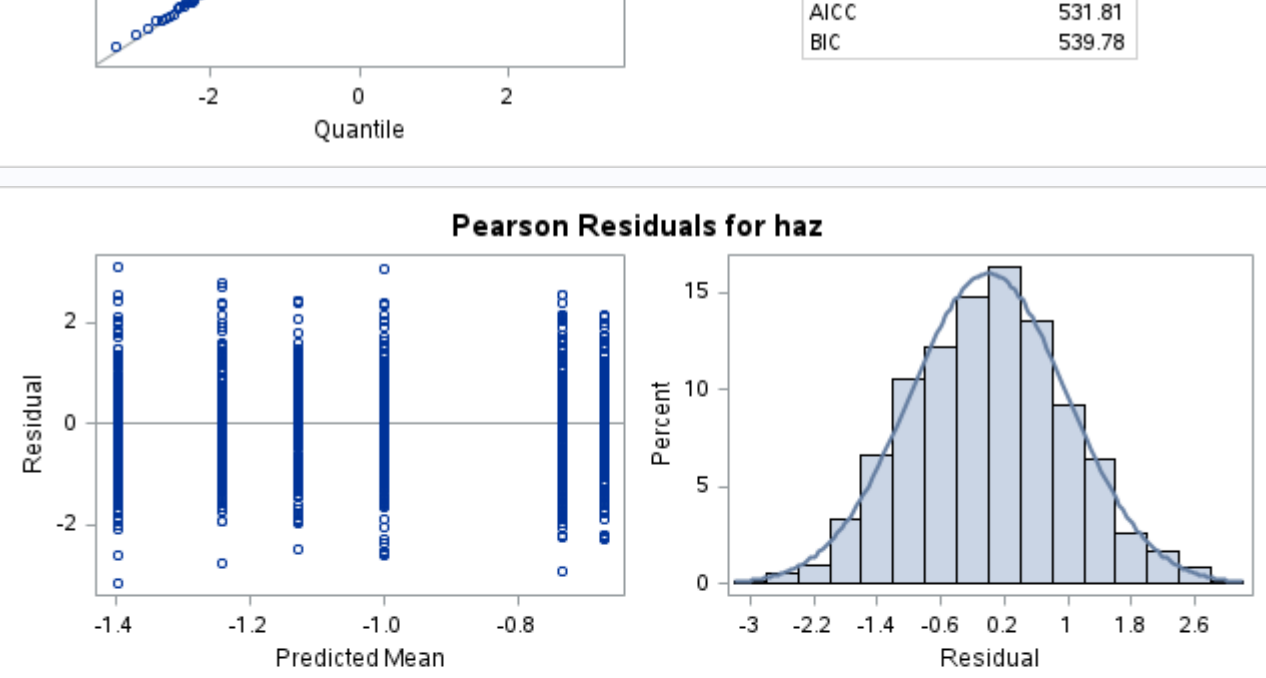
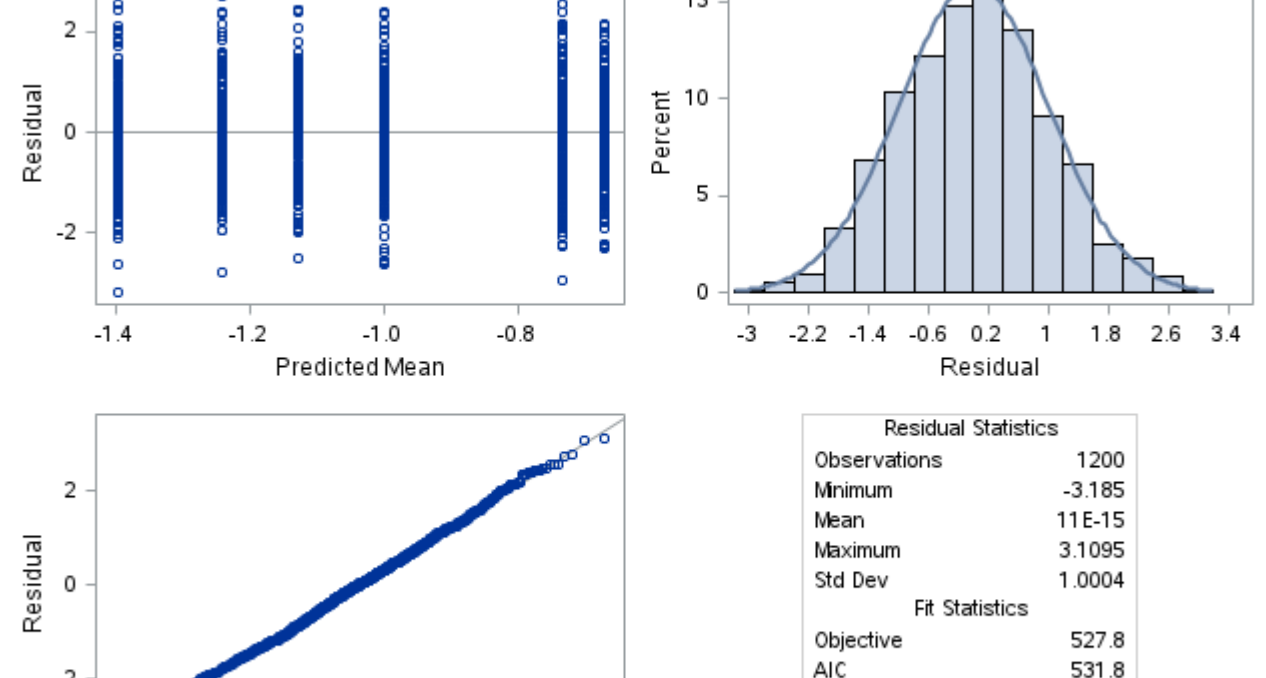
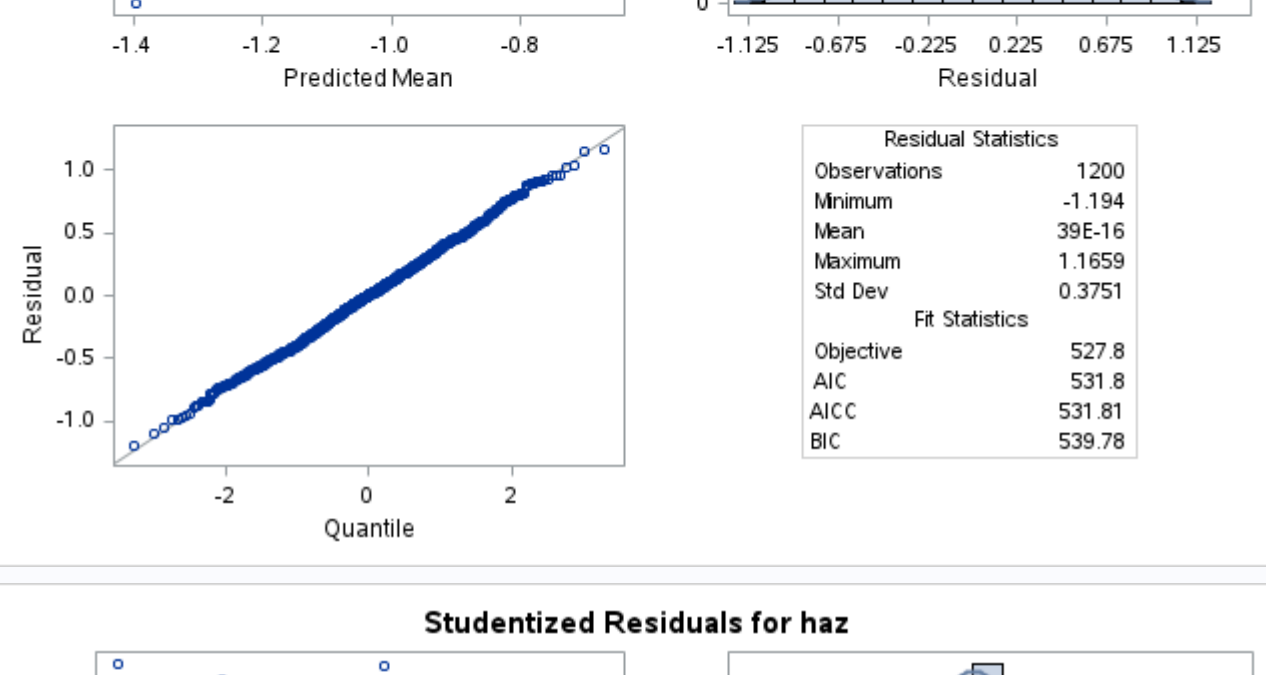
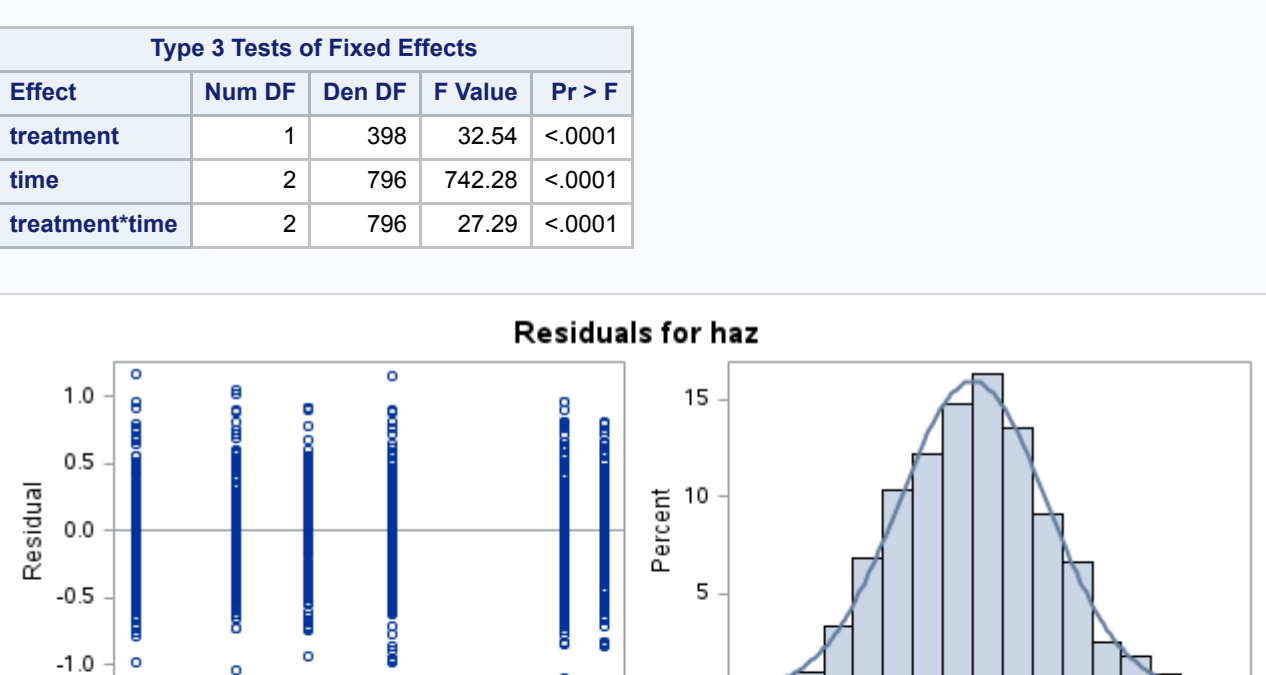
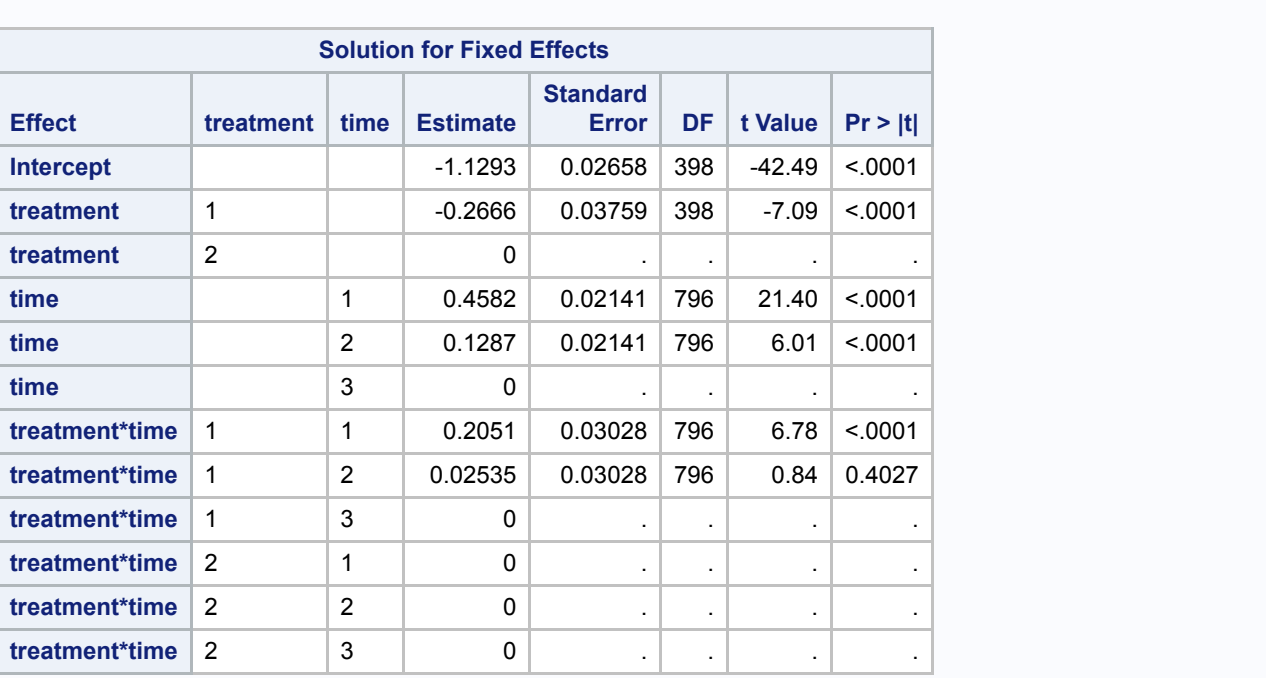
Fit Statistics	
-2 Res Log Likelihood	527.8
AIC (Smaller is Better)	531.8
AICC (Smaller is Better)	531.8
BIC (Smaller is Better)	539.8

Null Model Likelihood Ratio Test	
DF	Chi-Square
1	555.84
	<.0001

Solution for Fixed Effects	
Effect	treatment
Estimate	-1.1293
Standard Error	0.02658
DF	398
t Value	-42.49
Pr > t	<.0001
Intercept	1
Estimate	-0.2666
Standard Error	0.03759
DF	398
t Value	-7.09
Pr > t	<.0001
time	2
Estimate	0
Standard Error	-
DF	-
t Value	-
Pr > t	-
time	1
Estimate	0.4592
Standard Error	0.02141
DF	796
t Value	21.40
Pr > t	<.0001
time	2
Estimate	0.1287
Standard Error	0.02141
DF	796
t Value	6.01
Pr > t	<.0001
time	3
Estimate	0
Standard Error	-
DF	-
t Value	-
Pr > t	-
treatment*time	1
1	0.2051
2	0.03028
3	0.03028
796	6.78
<.0001	
treatment*time	1
2	0.02535
3	0.03028
796	0.84
0.4027	
treatment*time	1
3	0
-	-
-	-
-	-
treatment*time	2
1	0
-	-
-	-
-	-
treatment*time	2
2	0
-	-
-	-
-	-
treatment*time	2
3	0
-	-
-	-
-	-

Type 3 Tests of Fixed Effects	
Effect	Num DF
treatment	1
time	2
treatment*time	2

Denom DF	F Value	Pr > F
1	396	32.54
2	796	742.28
2	796	27.29



Design and Analysis of Randomized Trials in Public Health
PUBH 526

Homework 11

You have designed an intervention intended to improve linear growth of infants in a resource-poor setting, and you will compare your intervention to standard of care in a randomized controlled trial using a completely randomized design. Infants (0-6 months old) will be assessed at baseline (prior to intervention), midline (during the intervention), and endline (after the intervention).

At each time point, you will measure their length (height measured while lying on a flat surface) and use that to calculate their height-for-age Z-score (HAZ). HAZ is a measure that compares a child's height to the average height of healthy children of the same age and sex, and the number is specifically the number of standard deviations a child is above or below the healthy age- and sex-standardized average. A HAZ of -1, for example, means a child is one standard deviation shorter than a healthy child of their age and sex, and a HAZ < -2 leads to a child being classified as stunted, which indicates chronic undernutrition.

The dataset "growth" has HAZ data for 400 infants that were randomized to receive either the standard of care (treatment=1) or your new intervention (treatment=2). Each child ("ID") is measured at the three time points described above (time=1 for baseline, time=2 for midline, and time=3 for endline). Analyze these data using a longitudinal model (option 4 in Topic 14). What effect did the intervention have? Show your process, from exploring your data to formulating your hypotheses to thoroughly answering this question, and include your SAS code. (10 points)

My Process – Note: I check the log after running SAS code for each step. I will use a 0.05 significance level to determine significance.

1. I first looked at the dataset loaded in SAS to understand how it is set up and make sure there are no obvious errors in data entry. I did this both to get an idea of what names each variable has, how the table is formatted (ex. long versus wide), etc., as well as to give a first pass through to catch really obvious errors in the data set and how it was read by SAS.
 - a. From this step, I was able to tell that there were no initially obvious errors in the data set that would keep me from going any further (however it is pretty long, so I definitely need to double check this with code). I also could see that there are four columns initially sorted by (participant) ID then time. It is in long format with three rows per participant (one for each time period). ID is every integer from 1 to 400; treatment is 1 or 2; time is 1, 2, or 3; and haz has two decimal places.
2. I then used proc contents to see total number of observations and variables to confirm that it is as expected. I also looked at the variable type for each variable for my own reference.
 - a. This step showed there were 1200 observations which makes sense given there should be 3 observations for each of 400 infants [Output 2c1]. As expected, there are 4 variables (ID, haz, time, treatment), all of which are numeric [Output 2c2].

```

1 libname hw11 '/home/u63555875/PUBH526';
2
3 data growth; set hw11.growth; run;
4
5 proc contents data = growth;
6 run;
7

```

b. Code:

Observations	1200
Variables	4
Indexes	0
Observation Length	32
Deleted Observations	0

Alphabetic List of Variables and Attributes			
#	Variable	Type	Len
1	ID	Num	8
4	haz	Num	8
3	time	Num	8
2	treatment	Num	8

c. Relevant Output: 1-

2-

3. Then I used proc freq with a tables statement to check that there are 3 observations per person (1 per period, all within 1 treatment), 40 observations for each time period, and 600 observations for each treatment in the data set.
 - a. This step verified that the table does not appear to be missing any data or have issues with people having missing data or too many observations for one participant, etc.

```

8 proc freq data = growth;
9 tables id*time id*treatment/nocum norow nocol nopercnt;
10 run;

```

b. Code:

Table of ID by time					
ID	time			Total	
	1	2	3		
1	1	1	1	3	
2	1	1	1	3	
3	1	1	1	3	
4	1	1	1	3	
5	1	1	1	3	
6	1	1	1	3	
7	1	1	1	3	

Table of ID by treatment			
ID	treatment		
	1	2	Total
1	3	0	3
2	3	0	3
3	3	0	3
4	3	0	3
5	3	0	3
6	3	0	3
7	3	0	3

c. Example of the output:

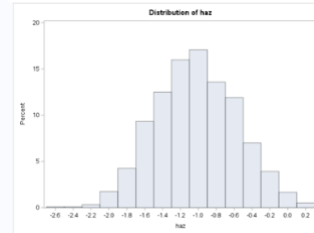
4. Then, I used proc univariate to get a basic idea of the data (mean, min, max, standard deviation) and ensure these values are reasonable considering the outcome (height-for-age Z-score). I also used this to generate a histogram to get a visual sense of the distribution of HAZ values overall.
- The mean haz was -1.03, with a minimum value of -2.59, a maximum value of 0.230, and a standard deviation of 0.46 minutes [see Relevant Output]. Based on what I know of HAZ (measure of size relative to standard value) and the study population (infants in a resource poor region), these values seem reasonable. The histogram also shows that almost all haz values are below zero which seems logical for the resource poor setting where undernutrition may be more prevalent. This also gives me an initial idea of what range of values to expect when exploring my data further and the potential differences across different groups.

```
12 proc univariate data = growth;
13 var haz;
14 histogram haz;
15 run;
```

b. Code:

Mom	
N	1200
Mean	-1.028525
Std Deviation	0.45667936
Skewness	0.04035143
Uncorrected SS	1519.4951
Coeff Variation	-44.401386

Quantiles (Definition 5)	
Level	Quantile
100% Max	0.230
99%	0.035
95%	-0.240
90%	-0.440
75% Q3	-0.705
50% Median	-1.030
25% Q1	-1.350
10%	-1.600
5%	-1.785
1%	-2.015
0% Min	-2.590



c. Relevant Output:

5. Next, I started exploring how the values generally compared across the six groups with different combinations of treatment and time period (SOC in period 1, SOC in period 2, SOC in period 3, NEW in period 1, NEW in period 2, NEW in period 3). Because I wanted to use a by statement in proc means to make this comparison, I first had to use proc sort to sort by treatment & time. Then, I used proc means with a by statement for treatment & time and a var statement specifying haz to get the n, mean, std dev, min, and max values for the value of haz in each of the six treatment/time period combinations.

- a. Across each of the six groups, the standard deviation was roughly the same (the lowest was 0.35 and the highest was 0.40), especially when considering the values expected for haz. Based on this initial summary of the data, it looks like treatment 1 (the standard of care) consistently produces lower (more negative) average haz across all time periods than treatment 2 (the new intervention), but the difference between treatment 1 and 2 on average is greater in later time periods. The difference between average haz looks much larger for treatment 2 than treatment 1 at time 3, but it looks pretty minimal at time 1—statistical tests must be done, though, to determine the significance of these differences. If significance tests reflect these same trends, it suggests that there is a treatment*time effect and the intervention improved haz scores for infants relative to the control group. We will have to see.
- b. I also graphed these differences which helps to visually show these differences in group means.

c. Code:

```

17 proc sort data = growth;
18 by treatment time;
19 run;
20
21 proc means data = growth;
22 by treatment time;
23 var haz;
24 output out=out1 mean=ave;
25 run;
26
27 symbol1 v=dot i=join c=black;
28 symbol2 v=dot i=join c=red;
29 proc gplot data=out1;
30 plot ave*time=treatment;
31 run;

```

c. Code:

The MEANS Procedure

treatment=1 time=1				
Analysis Variable : haz				
N	Mean	Std Dev	Minimum	Maximum
200	-0.7325500	0.4012919	-1.8300000	0.2300000

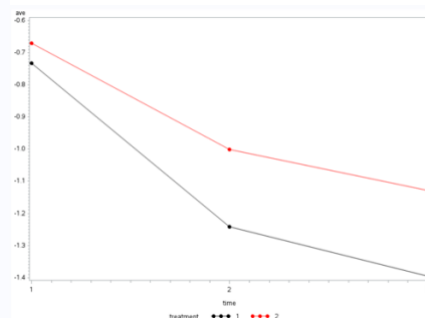
treatment=1 time=2				
Analysis Variable : haz				
N	Mean	Std Dev	Minimum	Maximum
200	-1.2418500	0.3858405	-2.2900000	-0.2000000

treatment=1 time=3				
Analysis Variable : haz				
N	Mean	Std Dev	Minimum	Maximum
200	-1.3868000	0.3836917	-2.5900000	-0.2300000

treatment=2 time=1				
Analysis Variable : haz				
N	Mean	Std Dev	Minimum	Maximum
200	-0.6710500	0.3535184	-1.5400000	0.1300000

treatment=2 time=2				
Analysis Variable : haz				
N	Mean	Std Dev	Minimum	Maximum
200	-1.0005000	0.3863657	-1.9900000	0.1500000

treatment=2 time=3				
Analysis Variable : haz				
N	Mean	Std Dev	Minimum	Maximum
200	-1.1262500	0.3493655	-2.0700000	-0.2100000



d. Relevant Output:

6. Now that I have confirmed that the table has the data entered as I would expect with no clear errors and I have gotten an approximate idea of the values of the data overall, as well as within each group, I will use proc mixed to create a model. The model code will have id, time, & treatment as class variables because they are categorical; a model statement that models haz as the outcome and time, treatment, and treatment*time as predictors; and a repeated statement to specify that the time period is repeated for each person (subject=id). I will try it with both a compound symmetry and a first order autoregressive covariance structure to see if they create any differences and which performs better based on fit statistics [I am assuming, though, that the covariance structure is the same for each participant and that variance does not changed between time periods].

- Proc mixed information output is as expected (model/class/etc. information)
- Log is happy and model converges for both covariance structures

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	1083.63138550	
1	1	527.79573624	0.00000000

Convergence criteria met.

- Output for convergence (CS cov):

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	1083.63138550	
1	2	553.70498089	0.00000000

Convergence criteria met.

- Output for convergence (AR1 cov):

- Comparing models with difference covariance structures: CS -> AIC=531.8, AR(1)->AIC=557.7 [Smaller is better, so I will use CS for the remainder of this data analysis]

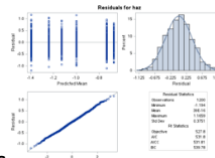
Fit Statistics	
-2 Res Log Likelihood	527.8
AIC (Smaller is Better)	531.8
AICC (Smaller is Better)	531.8
BIC (Smaller is Better)	539.8

- Output for CS cov:

Fit Statistics	
-2 Res Log Likelihood	553.7
AIC (Smaller is Better)	557.7
AICC (Smaller is Better)	557.7
BIC (Smaller is Better)	565.7

- Output for AR1 cov:

- Model Assumptions (constant variance, normality of residuals) Satisfied: Predicted Mean vs. Residual plot [top left] shows no clear pattern nor significant outliers, Q-Q plot [bottom left] is almost perfectly linear



- Output for model assumptions

- Time*treatment (treatment effect) is significant ($p < 0.0001$) which indicates there is a significant intervention effect.

```

33 proc mixed data = growth plots = all;
34   class id time treatment;
35   model haz = treatment time treatment*time / ddfm=kr;
36   repeated time / subject = id type = cs r = 1;
37 run;
38

```

- Code:

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
treatment	1	398	32.54	<.0001
time	2	796	742.28	<.0001
time*treatment	2	796	27.29	<.0001

- Relevant Output:

7. Finally, I added LSMeans estimation and I used slice to run the F-test to compare the treatments at each time point.

- a. The slice-wise F-test shows that the treatments were not significantly different from each other at time point 1 ($p > 0.05$), but they were at time point 2 and time point 3. Building on this with the difference estimates, the mean HAZ for treatment 1 (SOC) relative to the mean HAZ for treatment 2 (intervention) at time 2 was on average -0.2413 and at time 3 was on average -0.2666. This means that at both time 2 (midline) and time 3 (endline) there was a statistically significant improvement in mean HAZ scores for infants who received the intervention relative to those who received the standard of care, estimated at a group average HAZ that was 0.2413 and 0.2666 higher for the intervention group compared to the standard of care group at midline and baseline, respectively.

```
33 proc mixed data = growth plots = all;
34   class id time treatment;
35   model haz = treatment time treatment*time / ddfm=kr;
36   repeated time / subject = id type = cs r = 1;
37   lsmeans treatment*time / slice = time adjust=tukey;
38 run;
```

- b. Code:

Tests of Effect Slices					
Effect	time	Num DF	Den DF	F Value	Pr > F
time*treatment	1	1	624	2.68	0.1023
time*treatment	2	1	624	41.21	<.0001
time*treatment	3	1	624	50.32	<.0001

- c. Relevant Output Slice:

Differences of Least Squares Means											
Effect	time	treatment	_time	_treatment	Estimate	Standard Error	DF	t Value	Pr > t	Adjustment	Adj P
time*treatment	1	1	1	2	-0.06150	0.03759	624	-1.64	0.1023	Tukey-Kramer	0.5747
time*treatment	1	1	2	1	0.5093	0.02141	796	23.79	<.0001	Tukey-Kramer	<.0001
time*treatment	1	1	2	2	0.2680	0.03759	624	7.13	<.0001	Tukey-Kramer	<.0001
time*treatment	1	1	3	1	0.6633	0.02141	796	30.98	<.0001	Tukey-Kramer	<.0001
time*treatment	1	1	3	2	0.3967	0.03759	624	10.55	<.0001	Tukey-Kramer	<.0001
time*treatment	1	2	2	1	0.5708	0.03759	624	15.19	<.0001	Tukey-Kramer	<.0001
time*treatment	1	2	2	2	0.3295	0.02141	796	15.39	<.0001	Tukey-Kramer	<.0001
time*treatment	1	2	3	1	0.7248	0.03759	624	19.28	<.0001	Tukey-Kramer	<.0001
time*treatment	1	2	3	2	0.4582	0.02141	796	21.40	<.0001	Tukey-Kramer	<.0001
time*treatment	2	1	2	2	-0.2413	0.03759	624	-6.42	<.0001	Tukey-Kramer	<.0001
time*treatment	2	1	3	1	0.1540	0.02141	796	7.20	<.0001	Tukey-Kramer	<.0001
time*treatment	2	1	3	2	-0.1126	0.03759	624	-3.00	0.0028	Tukey-Kramer	0.0335
time*treatment	2	2	3	1	0.3963	0.03759	624	10.52	<.0001	Tukey-Kramer	<.0001
time*treatment	2	2	3	2	0.1287	0.02141	796	6.01	<.0001	Tukey-Kramer	<.0001
time*treatment	3	1	3	2	-0.2666	0.03759	624	-7.09	<.0001	Tukey-Kramer	<.0001

- d. Relevant Output Estimates:

Note: We only care about the pairwise comparisons within this table that are at the same time point (time 1 treatment 1 vs. time 1 treatment 2; time 2 treatment 1 vs. time 2 treatment 2; time 3 treatment 1 vs. time 3 treatment 3)

In conclusion, the intervention did, on average, result in statistically significant improvements in HAZ (closer to the healthy standardized size for their sex and age) compared to the standard of care at midline and endline. The average improvement was estimated at 0.2413 (midline) and 0.2666 (endline) closer to 0 (the healthy standardized size based on sex & age). While this difference was significant, more expertise in the field would be required to say what impact this quarter of a standard deviation improvement in size toward the healthy standard has for infant health and wellbeing.